The sodium atom is in the first excited state.

Question 2

Calculate the wavelength of the photon of energy emitted as the excited atom returns to the ground state.

($h = 4.14 \times 10^{-15}$ eV s, $c = 3.0 \times 10^{8}$ m s^{-1})

m

3 marks

An electron gun accelerates electrons across a potential difference of 2500 V. The initial speed of the electrons can be considered to be almost zero.

Question 3

Show that the final speed of the electrons is approximately 10% of the speed of light.

($m_e = 9.1 \times 10^{-31}$ kg, $c = 3.0 \times 10^{8}$ m s^{-1}, $e = 1.6 \times 10^{-19}$ C)

4 marks

Katie and Jane are discussing wave-particle duality. Jane wonders whether wave-particle duality might explain why she missed hitting the softball in a recent match – maybe the wave nature of the softball allowed it to diffract around the bat! Katie said that this was not a reasonable explanation and that we cannot see the wave nature of a softball.

A softball has a mass of 0.20 kg and the pitcher throws it at about 30 m s^{-1}.

Question 4

Explain to Jane, using an appropriate calculation, why she would be unable to see the wave nature of a moving softball. ($h = 6.63 \times 10^{-34}$ J s)

3 marks

In an experiment to demonstrate the photoelectric effect, physics students allow light of various frequencies to fall on a metal surface in a photocell. The photoelectrons are decelerated across a retarding voltage, and the stopping potential, V_s, is measured for each frequency. The data they obtained is graphed in Figure 2.

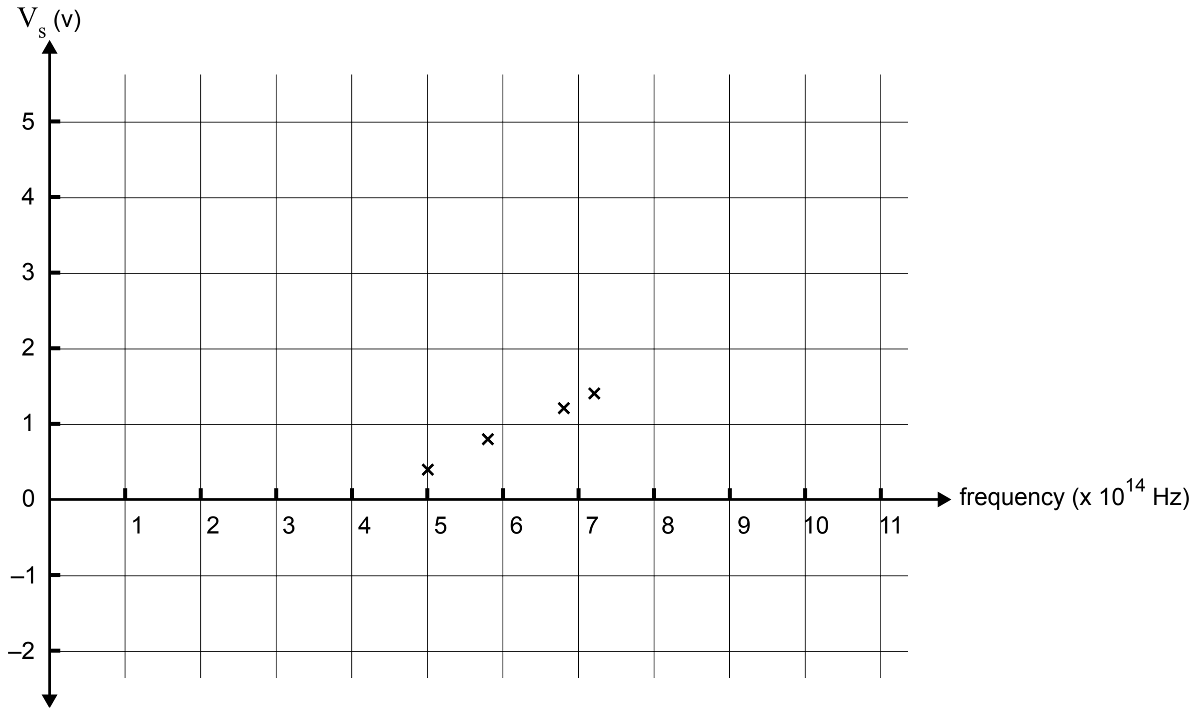

Figure 2

The students use the data points on the graph to determine a value for the work function of the metal.

Question 5

Determine the magnitude and unit of the work function for this metal surface.

Magnitude	Unit

2 marks

Question 6

What is the maximum kinetic energy (in eV) of the photoelectrons produced when ultraviolet light of frequency 1.93×10^{16} Hz is incident on the metal surface? ($h = 4.14 \times 10^{-15}$ eV s)

eV

3 marks

END OF QUESTION AND ANSWER BOOK